

W03

Algebra for Honor Mathematics

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Contents

1	Q1	2
2	Q2	3
3	Q3	4
4	Q4	5
5	Q5	7
6	Q6	11

§1 Q1

(a)

$$\begin{aligned}\overline{G} &= \{3, 5, 7, 8, 10\} \\ \overline{G} \cap H &= \{5, 7\}.\end{aligned}$$

(b)

$$\begin{aligned}F \cup H &= \{1, 4, 5, 7, 9, 10\} \\ G \cap (F \cup H) &= \{1, 4, 9\}\end{aligned}$$

(c)

$$\begin{aligned}G \cup H &= \{1, 2, 4, 5, 6, 7, 9\} \\ \overline{(G \cup H)} &= \{3, 8, 10\}\end{aligned}$$

(d)

$$\begin{aligned}(F \cap H) &= \{4, 5, 9\} \\ (F \cap H) - G &= \{5\}\end{aligned}$$

§2 Q2

Prove the following statement.

$$\forall x \in \mathbb{N}, \exists y \in \mathbb{N}, (x^2 + 2 - y) \mid (10 - x^2 - y^2)$$

Solution. Let x be arbitrary number in $\mathbb{N}$. We select y to be $x^2 + 1$ in $\mathbb{N}$. substituting $y = x^2 + 1$ into the expression, we

$$(x^2 + 2 - x^2 - 1) \mid (10 - x^2 - (x^2 + 1)^2)$$

This implies

$$(1) \mid (10 - x^2 - x^4 - 2x^2 - 1),$$

which equals to

$$(1) \mid (-x^4 - 3x^2 + 9)$$

there exists a number $k = -x^4 - 3x^2 + 9$ in $\mathbb{Z}$ such that

$$(-x^4 - 3x^2 + 9) \cdot 1 = -x^4 - 3x^2 + 9$$

for all $x \in \mathbb{N}$.

Thus, according to the definition of divisibility,

$$\forall x \in \mathbb{N}, \exists y \in \mathbb{N}, (x^2 + 2 - y) \mid (10 - x^2 - y^2).$$

■

§3 Q3

Prove the following statement.

Solution. Let p, q, r be arbitrary in $\mathbb{Z}$.

By Elimination, it suffices to prove that

$$[(p^2q \mid (r^2 - q)) \wedge (pq^3 \mid qr)] \implies (pq \mid (q(7 + 5r) - 7r^2))$$

Suppose $(p^2q \mid (r^2 - q))$ and $(pq^3 \mid qr)$, by the definition of divisibility, there exists $a, b \in \mathbb{Z}$ such that

$$\begin{aligned} r^2 - q &= p^2qa \\ qr &= pq^3b. \end{aligned}$$

Moreover, we expand $q(7 + 5r) - 7r^2$, we get

$$7q + 5qr - 7r^2.$$

Rerarrange and factoring -7 out of $-7r^2 + 7q$ we obtains

$$-7r^2 + 7q + 5qr = -7(r^2 - q) + 5qr.$$

Subtitute $r^2 - q = p^2qa$ and $qr = pq^3b$ in, we get

$$-7p^2qa + 5pq^3b.$$

Factoring pq out, we get

$$pq(-7pa + 5q^2b).$$

Since $p, q, r, a, b \in \mathbb{Z}$, therefore

$$-7pa + 5q^2b \in \mathbb{Z}$$

such that

$$pq(-7pa + 5q^2b) = q(7 + 5r) - 7r^2.$$

Thus, by the definition of divisibility, we obtain

$$pq \mid (q(7 + 5r) - 7r^2.)$$

■

§4 Q4

(a) **Solution.** This statement is true.

Suppose x be arbitrary in $\mathcal{F}$. By the definition of set, we have

$$42 \mid x + 18, x \in \mathbb{Z}.$$

By the definition of divisibility, there exists $k \in \mathbb{Z}$ such that

$$42k = x + 18.$$

Rearrange, we get

$$x = 42k - 18.$$

Factor 3 out, we get

$$x = 3(14k - 6).$$

Since $k \in \mathbb{Z}$,

$$x = 3(14k - 6), 14k - 6 \in \mathbb{Z}.$$

This yields $x \in \mathcal{K}$.

Moreover, substitute $x = 42k - 18$ into $x + 18$, we get

$$42k - 18 + 18 = 42k.$$

This is equal to

$$7 \cdot (6k).$$

Since $k \in \mathbb{Z}$, $6k \in \mathbb{Z}$.

Therefore, there exists $6k \in \mathbb{Z}$ such that

$$x + 18 = 7 \cdot 6k.$$

By the definition of divisibility, we have

$$7 \mid x + 18.$$

We obtain $x \in \mathcal{P}$.

Therefore, we proved

$$\forall x \in \mathbb{Z}, [(x \in \mathcal{F}) \implies (x \in (\mathcal{K} \cap \mathcal{P}))]$$

■

(b) **Solution.** This statement is false.

We disprove this statement by giving a counterexample. Let $x = 45 \in \mathbb{Z}$.

Since

$$45 + 18 = 63$$

and

$$42 \nmid 63.$$

We have

$$45 \notin \mathcal{F}$$

However, $3 \mid 45$. So

$$45 \in \mathcal{K}.$$

Moreover,

$$45 + 18 = 63.$$

and

$$7 \mid 63.$$

So

$$45 \in \mathcal{P}.$$

Thus, we proved that

$$\exists x \in \mathbb{Z}, [(x \notin \mathcal{F}) \wedge (x \in (\mathcal{K} \cap \mathcal{P}))]$$

, which is a negation of the original statement. So, we disprove the original statement. ■

§5 Q5

- (a) **Solution.** In order to prove that $K \cap L \neq \emptyset$. We need to show that there exists x such that $x \in K$ and $x \in L$. By the definition of sets K and L , it suffices to prove that

$$\exists x, s, u \in \mathbb{Z}, 3000 \leq s \leq 4000, x = (2s)^2 = 2u^3.$$

We pick $s = 2 \cdot (12)^3$ and $u = 2 \cdot 12^2$

First, we need to prove that $3000 \leq s \leq 4000$.

$$s = 2 \cdot 1728 = 3456,$$

which satisfies $3000 \leq 3456 \leq 4000$.

Next, we need to prove $(2s)^2 = 2u^3$. Substituting in $s = 2 \cdot (12)^3$ and $u = 2 \cdot 12^2$, we have

$$(2s)^2 = [2 \cdot (2 \cdot 12^3)]^2 = 2^4 \cdot 12^6 = 2(2^3 \cdot 12^6) = 2(2 \cdot 12^2)^3 = 2u^3.$$

Since $x = (2s)^2$ with $s \in \mathbb{Z}$ and $3000 \leq s \leq 4000$, we have $x \in K$; moreover, since $x = 2u^3$ with $u \in \mathbb{Z}$, we have $x \in L$.

Thus,

$$x \in (K \cap L)$$

Hence, we prove that sets K and L are not disjoint. ■

(b) **Solution.** The following statement is true.

In order to prove the statement, by definition of the sets, we need to show that there is exactly one pair of integers s, u such that

$$3000 \leq s \leq 4000, (2s)^2 = 2u^3.$$

Assume $(2s)^2 = 2u^3$. We have

$$\begin{aligned} (2s)^2 = 2u^3 &\implies 4s^2 = 2u^3 \\ &\implies 2s^2 = u^3. \end{aligned}$$

According to the lemma 2 shown after the solution for Question (b), there exists $k \in \mathbb{Z}$ such that

$$u = 2k.$$

Substituting this back into $2s^2 = u^3$, we get

$$\begin{aligned} 2s^2 = (2k)^3 &\implies 2s^2 = 8k^3 \\ &\implies \frac{1}{4}s^2 = k^3 \\ &\implies \left(\frac{1}{2}s\right)^2 = k^3. \end{aligned}$$

Since $s^2 = 2(2k^3)$, by the lemma 1, s is even, so $\frac{s}{2} \in \mathbb{Z}$. By the proposition given for this question, k is a perfect square. Therefore, there exists some non-negative $t \in \mathbb{Z}$ such that

$$k = t^2.$$

Substituting $k = t^2$ into the equation, we get

$$\begin{aligned} \left(\frac{1}{2}s\right)^2 &= (t^2)^3 \\ &= t^6. \end{aligned}$$

Since s and t are both non-negative, taking the square root of both sides gives

$$\begin{aligned} \frac{1}{2}s &= t^3 \\ \implies s &= 2t^3. \end{aligned}$$

In order for $(2s)^2 \in K$, we must have

$$3000 \leq s \leq 4000.$$

Substituting $s = 2t^3$ in,

$$\begin{aligned} 3000 &\leq 2t^3 \leq 4000, \\ \implies 1500 &\leq t^3 \leq 2000. \end{aligned}$$

Try the first case,

$$11^3 = 1331 < 1500$$

and the second case

$$13^3 = 2197 > 2000.$$

There is only one $t \in \mathbb{N}$ satisfies

$$1500 \leq t^3 \leq 2000,$$

which is

$$t = 12.$$

Therefore,

$$s = 2(12)^3 = 3456$$

and

$$u = 2k = 2t^2 = 2(12)^2 = 288.$$

Thus, the only element in both sets K and L is

$$x = (2s)^2 = (2 \cdot 3456)^2 = 2(288)^3.$$

it follows that

$$|K \cap L| = 1.$$

Therefore, the statement is true. ■

Lemma 1.

$$\forall n \in \mathbb{Z}, [(2 \mid n^2) \implies (2 \mid n)].$$

Proof. Suppose n is arbitrary in $\mathbb{Z}$, for contrapositive, $2 \nmid n$.

By the definition of divisibility, there exists $k \in \mathbb{Z}$ such that

$$n = 2k + 1.$$

This implies

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1$$

Since $2k^2 + 2k \in \mathbb{Z}$, we have

$$n^2 = 2(2k^2 + 2k) + 1$$

By the definition of divisibility, we get

$$2 \nmid n^2.$$

Thus, we prove that if $2 \mid n^2$, then $2 \mid n$. ■

Lemma 2.

$$\forall n \in \mathbb{Z}, [(2 \mid n^3) \implies (2 \mid n)].$$

Proof. Suppose n is arbitray in $\mathbb{Z}$, for contrapositive, $2 \nmid n$.

By the definition of divisibility, there exists $k \in \mathbb{Z}$ such that

$$n = 2k + 1.$$

This implies

$$n^3 = (2k + 1)^3 = 8k^3 + 12k^2 + 6k + 1.$$

Since $4k^3 + 6k^2 + 3k \in \mathbb{Z}$, we have

$$n^3 = 2(4k^3 + 6k^2 + 3k) + 1.$$

By the definition of divisibility, we get

$$2 \nmid n^3.$$

Thus, we prove that if $2 \mid n^3$, then $2 \mid n$. ■

§6 Q6

1.

$$S = \mathcal{E}, \quad P(x, y) = P_2(x, y).$$

2.

$$S = \mathcal{D}, \quad P(x, y) = P_4(x, y).$$

3.

$$S = \mathcal{M}, \quad P(x, y) = P_1(x, y).$$

4.

$$S = \mathcal{J}, \quad P(x, y) = P_3(x, y).$$